Detailed Evaluation of Ecosystem Processes (DEEP) Field Worksheet

Western Gulf Coastal Plain reference assessment | western-gulf-coastal-plain@1
USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional

Reach ID: NHDPlusID 10000100012345 **Reach Length:** 1,000 ft **Date:** _____ **Assessor(s):** _____ **Coordinates:** 43.68581, -72.23671
Site: Mink Brook | drainage area 12.43 km2 | HR reach watershed (STAF site engine v0.5.0)

RECORDING INSTRUCTIONS

Record the measured value of each metric in its units. Metrics measured in the field are marked F and metrics answered from the desk are marked D. A value printed in the Value cell was already answered from the desk and carries its source. Enter the values in DEEP, which reads each one on the metric's reference curve to give an index from 0 to 1. The function score is the mean index of the function's metrics times 15 and reads Functioning (15 to 11), Functioning At-Risk (10 to 6), or Non-Functioning (5 to 0). A metric that does not apply is marked NA and drops out of the score. Where a metric lists curve sets, mark the set that fits the reach.

Function	Metrics	Value	Index
HYDROLOGY FUNCTIONS			
Catchment hydrology: Runoff and infiltration sustain natural flow regime, carry appropriate sediment and nutrients from uplands, and reliably cue spawning/migration of aquatic life.	**Impervious surface** (%) D: Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. *DESKTOP: NLCD 2021*	**3.0**	
	Agriculture, crop and hay (%) D: Percent of the delineated watershed in cultivated crops plus hay and pasture, computed by DEEP. No field measurement. *DESKTOP: EPA StreamCat*	**12.75**	
Score:	**Notes/Other Metrics:**		
Surface water storage: Wetlands and storage features store floodwater, recharge groundwater, sustain baseflow, and provide low-velocity habitat.	**Wetland** (%) D: Percent of the watershed in woody plus herbaceous wetland (NLCD 2019), read from EPA StreamCat for the reach. No field measurement. *DESKTOP: Computed by DEEP (reference only)*	**0.4**	
Score:	**Notes/Other Metrics:**		
Reach inflow: Quantity + quality of inflow (tributaries, ditches, and pipes) does not provide harmful peaks or pollution, and supports diverse habitat conditions.	**Road density** (km/sq km) D: Road length per square kilometer of the delineated watershed, computed by DEEP. No field measurement. *DESKTOP: NLCD 2021*	**3.0**	
Score:	**Notes/Other Metrics:**		
Streamflow regime: Integrates the range of "typical" flows experienced by other processes. Examines the degree to which upstream infrastructure or land uses have fundamentally altered flow regimes (e.g., hydropeaking, dams, withdrawals).	**Base flow index** (%) D: Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. *DESKTOP: EPA StreamCat*	**12.75**	
	Degree of regulation (%) D: Normal storage of upstream dams as a percent of mean annual runoff, both read from EPA StreamCat for the reach. No field measurement. *DESKTOP: Computed by DEEP*	**0.4**	
Score:	**Notes/Other Metrics:**		

HYDRAULICS FUNCTIONS			
Low flow and baseflow dynamics: Ensures habitat availability and water quality during low water levels, and indicates non-storm conditions experienced the majority of the time.	**Base flow index** (%) D: Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. *DESKTOP: EPA StreamCat*	**12.75**	
	Residual pool depth (cm) F: Measure thalweg depth at 10 or 15 evenly spaced stations between each pair of transects (100 or 150 per reach). Value: mean residual pool depth in cm, from that profile and the reach slope (NRSA physical habitat calculations).		
Score:	**Notes/Other Metrics:**		
High flow dynamics: Peak flows reshape channel without chronic instability, redistribute wood and sediment, reset habitats, and supply nutrients to floodplains.	**Impervious surface** (%) D: Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. *DESKTOP: NLCD 2021*	**3.0**	
Score:	**Notes/Other Metrics:**		
Floodplain connectivity: Frequent overbank flows reduce flood peaks, support riparian vegetation, create off-channel refugia, and extend nutrient processing time.	**Bankfull height above channel** (m) F: At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights.		
	Bankfull width:depth ratio F: At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).		
Score:	**Notes/Other Metrics:**		
Hyporheic connectivity: Surface water-groundwater exchange moderates temperature, transforms nutrients, supplies oxygen, and supports aquatic life.	**Embeddedness** (%) F: At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates.		
	Sand + fines not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
GEOMORPHOLOGY FUNCTIONS			
Channel evolution: Long-term changes in channel size and slope maintain dynamic equilibrium and diverse habitats while interacting with the floodplain.	**Bankfull height above channel** (m) F: At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights.		
	Bankfull width:depth ratio F: At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).		
Score:	**Notes/Other Metrics:**		
Channel and floodplain dynamics: Examines the role of bank processes and erosion in channel change. Observes patterns in sinuosity and curvature affecting habitat complexity.	**Sinuosity** F: Take a compass backsight bearing and the distance between each pair of neighboring transects. Value: channel length along those segments divided by the straight-line distance between the reach ends.		
	Bank angle (degrees) F: On both banks of each transect, lay a pole about 1 m long against the bank and read its angle from horizontal with a clinometer, in degrees. Value: mean of the 22 readings.		
Score:	**Notes/Other Metrics:**		

Sediment continuity: Balanced sediment supply and transport preserves bed elevations, substrate sizes, spawning/benthic habitats, and supports riparian succession.	**Relative bed stability** (log) F: Computed: the base 10 log of the geometric mean particle diameter divided by the largest diameter the bankfull flow can move, from the particle count, reach slope, bankfull depth, wood tally and residual pools (NRSA physical habitat calculations).		
	Sand + fines not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
Bed composition and bedform dynamics: Supports aquatic habitats through streambed material and bedform dynamics. Enhances habitat complexity and streambank stability.	**Embeddedness** (%) F: At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates.		
	Large wood volume (m3/100m) F: Between each pair of transects, tally every piece of wood at least 10 cm in diameter and 1.5 m long in or above the bankfull channel, by length and diameter class. Value: wood volume per 100 m of channel, from the tally.		
	Substrate size not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
PHYSICOCHEMISTRY FUNCTIONS			
Light & thermal regime: Shade, solar input, and groundwater keep temperature and light within natural ranges for stream metabolism and dissolved oxygen.	**Canopy density, mid-channel** (%) F: At mid-channel on each transect, read a convex spherical densiometer facing upstream, downstream, left and right (covered points of 17). Value: mean of the 44 readings, as a percent. Curve set: [ ] All streams (pooled) [ ] Small stream (10 to 100 km2) [ ] Large stream (over 100 km2)		
	Chlorophyll a not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
Carbon processing: Organic matter is captured and broken down, fueling food webs, balancing production/ respiration, moderating pH/ redox, and supplying nutrients.	**Riparian veg cover, woody** (frac) F: In a 10 m by 10 m plot on each bank at each transect, estimate the areal cover class of woody vegetation in the canopy, understory and ground layer. Value: summed woody cover of the 3 layers, averaged over the 22 plots, as a fraction.		
Score:	**Notes/Other Metrics:**		
Nutrient cycling: Nitrogen and phosphorus transformations feed primary production, regulate excess loads, and store or release nutrients.	**Total nitrogen** (mg N/L) F: Total nitrogen (mg N/L), lab analysis of the unfiltered mid-channel grab sample from the X-site, kept on ice.		
	Total phosphorus (ug/L) F: Total phosphorus (ug/L), lab analysis of the mid-channel grab sample from the X-site, kept on ice.		
Score:	**Notes/Other Metrics:**		
Water & soil quality: Physical and chemical processes limit pollutants, retain beneficial constituents, and protect ecological and human uses.	**Specific conductivity** (uS/cm) F: Specific conductance (uS/cm). NRSA reports the lab value of a mid-channel grab sample at the X-site (the reach center); a calibrated multi-parameter meter reads the same quantity in the stream there.		
	pH not scored *Insufficient reference support. No value is needed.*	**NA**	
	Turbidity not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		

BIOLOGY FUNCTIONS			
Habitat provision: Channel and floodplain structure supply depth, velocity, substrate diversity, and vegetation to support native organisms through all life stages.	**Natural fish cover** (frac) F: In the 10 m of channel centered on each transect, estimate the areal cover class of each fish cover type on the form (wood, brush, overhanging vegetation, undercut banks, boulders, others). Value: summed cover of the natural types, averaged over the transects. Curve set: [ ] All streams (pooled) [ ] Small stream (10 to 100 km2) [ ] Large stream (over 100 km2)		
	Fast-water / riffle habitat not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
Population support: Physical and chemical conditions enable spawning, juvenile growth, migration, and stress survival of target taxa.	**Fish MMI** (MMI points) F: Computed: from the upstream electrofishing pass (every habitat between the transects, every fish identified and counted), score EPA's 8 regional fish metrics, several adjusted for watershed area, and rescale to 0 to 100 (NRSA 2018-19 technical support document).		
	Benthic total taxa richness not scored *Insufficient reference support. No value is needed.*	**NA**	
	Native fish taxa richness not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
Community dynamics: Species interactions sustain biodiversity, restrain invasives, and build resilience to disturbance.	**Benthic macroinvertebrate MMI** (MMI points) F: Computed: from the 11-transect composite D-frame sample (500 um mesh) and its fixed-count lab subsample, score EPA's 6 regional benthic metrics against the published floors and ceilings and rescale the sum to 0 to 100 (NRSA 2018-19 technical support document).		
	EPT taxa richness not scored *Insufficient reference support. No value is needed.*	**NA**	
	Shannon diversity (H') not scored *Insufficient reference support. No value is needed.*	**NA**	
	Tolerant individuals (%) ~ HBI not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
Watershed connectivity: Continuous pathways for free movement of water, sediment, and organisms supporting recolonization, maintaining species diversity, and enabling recovery.	**Mapped dams within one mile** (dams) D: Dams in the USACE National Inventory of Dams within 1 mile of the site, counted by DEEP. Note in the field any barrier the inventory does not show. *DESKTOP: EPA StreamCat*	**12.75**	
Score:	**Notes/Other Metrics:**		

METRICS NOT SCORED

Benthic total taxa richness, Chlorophyll a, EPT taxa richness, Fast-water / riffle habitat, Native fish taxa richness, Sand + fines, Shannon diversity (H'), Substrate size, Tolerant individuals (%) ~ HBI, Turbidity, pH. These metrics were considered for this assessment. These metrics are not scored. A metric withheld for insufficient reference support has no curve. No pool of least-disturbed stations in this ecoregion or its parents supports it, no national donor pool was shown to transfer, no modeled expectation passed validation, and no published criterion applies. No measurement is needed for the score. A value may still be recorded under Notes.